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Connector Assembly and Board-End Connector Thereof

Abstract

A connector assembly and board-end connector thereof, wherein the board-end connector comprises multiple terminal modules, each of which comprises a signal terminal and a metal shielding cover. The metal shielding cover provides a shielding environment for the signal terminal to reduce the level of crosstalk interference when the signal terminal transmits high-speed signals.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims the priority of China Patent Application No. 202410266313.5 filed on Mar. 8, 2024, in the State Intellectual Property Office of China, the disclosure of which is incorporated herein by reference.

BACKGROUND OF THE INVENTION

Field of the Invention

[0002] This application relates to a connector assembly and board-end connector thereof. More specifically, it refers to a connector assembly and board-end connector that can reduce crosstalk interference on signal terminals when transmitting high-speed signals.

Descriptions of the Related Art

[0003] Connector assemblies are widely used in various electronic devices. In recent years, connector assemblies have been designed to work with circuit boards. Generally, a connector assembly comprises a board-end connector and a mating component. The board-end connector comprises signal terminals, where the signal terminals can join a circuit board and mate with the board-end connector, thereby providing electrical signal transmission within electronic devices.

[0004] However, currently, signal terminals of board-end connectors are prone to crosstalk interference when transmitting high-speed signals, which causes interference and attenuation of the high-speed signal transmission. Therefore, how to suppress crosstalk interference so that the signal terminals of the board-end connector can achieve high-speed signal transmission is a problem that needs to be solved by those skilled in the art.

SUMMARY OF THE INVENTION

[0005] In view of the drawbacks of the prior art mentioned above, the present application provides a board-end connector, the board-end connector comprising: a connector insulating core, the connector insulating core comprising a major mating side, a major non-mating side, a major joining side, a major non-joining side, a connector insulating core mating space, and a connector insulating core joining space, wherein the major non-mating side and the major mating side are substantially different, and the major non-joining side and the major joining side are substantially different; a first major terminal module, the first major terminal module comprising a first major signal terminal, a first major metal shielding cover, and a first major signal terminal insulating core, the first major signal terminal comprising a first major signal terminal mating structure and a first major signal terminal joining structure, the first major metal shielding cover comprising a first major metal shielding cover mating shielding structure, a first major metal shielding cover joining shielding structure, a first major metal shielding cover mating passage, and a first major metal shielding cover joining passage, the first major signal terminal insulating core being mated with the first major signal terminal, and the first major terminal module being configured in the connector insulating core, such that the first major signal terminal mating structure is located in the connector insulating core mating space and the first major signal terminal joining structure is located in the connector insulating core joining space, the first major metal shielding cover being mated with the first major signal terminal insulating core, such that the first major metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the first major signal terminal mating structure to be mated via the first major metal shielding cover mating passage on the major mating side, and the first major metal shielding cover joining passage is in continuity with the connector insulating core joining space, allowing the first major signal terminal joining structure to be joined via the first major metal shielding cover joining passage on the major joining side, wherein the first major metal shielding cover mating shielding structure extends on the major non-mating side, and the first major metal shielding cover joining shielding structure extends on the major non-joining side; a second major terminal module, the second major terminal module comprising a second major signal terminal, a second major metal shielding cover, and a second major signal terminal insulating core, the second major signal terminal comprising a second major signal terminal mating structure and a second major signal terminal joining structure, the second major metal shielding cover comprising a second major metal shielding cover mating shielding structure, a second major metal shielding cover joining shielding structure, a second major metal shielding cover mating passage, and a second major metal shielding cover joining passage, the second major signal terminal insulating core being mated with the second major signal terminal, and the second major terminal module being configured in the connector insulating core, such that the second major signal terminal mating structure is located in the connector insulating core mating space and the second major signal terminal joining structure is located in the connector insulating core joining space, the second major metal shielding cover being mated with the second major signal terminal insulating core, such that the second major metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the second major signal terminal mating structure to be mated via the second major metal shielding cover mating passage on the major mating side, and the second major metal shielding cover joining passage is in continuity with the connector insulating core joining space, allowing the second major signal terminal joining structure to be joined via the second major metal shielding cover joining passage on the major joining side, wherein the second major metal shielding cover mating shielding structure extends on the major non-mating side, and the second major metal shielding cover joining shielding structure extends on the major non-joining side; and a major bridging conductive component, the major bridging conductive component comprising a first major bridging conductive component non-mating side shielding wall, the major bridging conductive component being configured in the connector insulating core, such that the first major bridging conductive component non-mating side shielding wall extends on the major non-mating side and is located between the first major terminal module and the second major terminal module, and the major bridging conductive component bridging over the first major signal terminal and the second major signal terminal, respectively joining the first major metal shielding cover and the second major metal shielding cover, allowing the first major bridging conductive component non-mating side shielding wall, the first major metal shielding cover mating shielding structure, and the second major metal shielding cover mating shielding structure to be electrically connected, providing a shielding environment for the first major signal terminal mating structure and the second major signal terminal mating structure, and allowing the first major metal shielding cover joining shielding structure and the second major metal shielding cover joining shielding structure to be electrically connected, providing a shielding environment for the first major signal terminal joining structure and the second major signal terminal joining structure.

[0006] Preferably, the board-end connector said above, wherein the first major terminal module and the second major terminal module are substantially the same; the connector insulating core comprises a first major terminal module core configuration structure and a second major terminal module core configuration structure, and the first major terminal module core configuration structure is configured to the first major terminal module, and the second major terminal module core configuration structure is configured to the second major terminal module.

[0007] Preferably, the board-end connector said above, further comprising: a third major terminal module, the third major terminal module comprising a third major signal terminal, a third major metal shielding cover, and a third major signal terminal insulating core, and the third major signal terminal comprises a third major signal terminal mating structure and a third major signal terminal joining structure, and the third major metal shielding cover comprises a third major metal shielding cover mating shielding structure, a third major metal shielding cover joining shielding structure, a third major metal shielding cover mating passage, and a third major metal shielding cover joining passage, and the third major signal terminal insulating core is mated with the third major signal terminal, and the third major terminal module is configured in the connector insulating core, positioning the third major signal terminal mating structure within the connector insulating core mating space, and positioning the third major signal terminal joining structure within the connector

insulating core joining space, and the third major metal shielding cover is mated with the third major signal terminal insulating core, so that the third major metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the third major signal terminal mating structure to mate via the third major metal shielding cover mating passage on the major mating side, and the third major metal shielding cover joining passage is in continuity with the connector insulating core joining space, allowing the third major signal terminal joining structure to join via the third major metal shielding cover joining passage on the major joining side, wherein the third major metal shielding cover mating shielding structure extends on the major non-mating side, and the third major metal shielding cover joining shielding structure extends on the major non-joining side; wherein the major bridging conductive component further comprises a second major bridging conductive component non-mating side shielding wall, and the second major bridging conductive component non-mating side shielding wall extends on the major non-mating side and is located between the second major terminal module and the third major terminal module. The major bridging conductive component also bridges across the second major signal terminal and the third major signal terminal, respectively joining the second major metal shielding cover and the third major metal shielding cover, thereby allowing the second major bridging conductive component non-mating side shielding wall, the second major metal shielding cover mating shielding structure, and the third major metal shielding cover mating shielding structure to be electrically connected, providing a shielding environment for the second major signal terminal mating structure and the third major signal terminal mating structure.
[0008] Preferably, the board-end connector said above, wherein the major bridging conductive component further comprises a major bridging conductive component mating side shielding wall and a major bridging conductive component non-mating side abutting wall; wherein the major bridging conductive component mating side shielding wall extends on the major mating side, providing a shielding environment for the first major signal terminal mating structure and the second major signal terminal mating structure; and the major bridging conductive component non-mating side abutting wall extends on the major non-mating side, respectively abutting the first major metal shielding cover and the second major metal shielding cover.
[0009] Preferably, the board-end connector said above, wherein the first major bridging conductive component non-mating side shielding wall is a bent type wall, allowing the first major bridging conductive component non-mating side shielding wall to elastically abut the first major metal shielding cover and the second major metal shielding cover, respectively.
[0010] Preferably, the board-end connector said above, wherein the first major bridging conductive component non-mating side shielding wall is a shear type wall, allowing the first major bridging conductive component non-mating side shielding wall to lock the first major metal shielding cover and the second major metal shielding cover.
[0011] Preferably, the board-end connector said above, wherein the connector insulating core further comprises a minor mating side, a minor non-mating side, a minor joining side, and a minor non-joining side, wherein the minor non-mating side and the minor mating side are substantially different, and the minor non-joining side and the minor joining side are substantially different; and the board-end connector further comprises: a first minor terminal module, and the first minor terminal module comprises a first minor signal terminal, a first minor metal shielding cover, and a first minor signal terminal insulating core, and the first minor signal terminal comprises a first minor signal terminal mating structure and a first minor signal terminal joining structure, and the first minor metal shielding cover comprises a first minor metal shielding cover mating shielding structure, a first minor metal shielding cover joining shielding structure, a first minor metal shielding cover mating passage, and a first major metal shielding cover joining passage, and the first minor signal terminal insulating core is mated with the first minor signal terminal, and the first minor terminal module is configured in the connector insulating core, so that the first minor signal terminal mating structure is located in the connector insulating core mating space and the first

minor signal terminal joining structure is located in the connector insulating core joining space, and the first minor metal shielding cover is mated with the first minor signal terminal insulating core, so that the first minor metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the first minor signal terminal mating structure to provide mating via the first minor metal shielding cover mating passage on the minor mating side, allowing the first major metal shielding cover joining passage to be in continuity with the connector insulating core joining space, allowing the first minor signal terminal joining structure to provide joining via the first major metal shielding cover joining passage on the minor joining side, wherein the first minor metal shielding cover mating shielding structure extends on the minor non-mating side, and the first minor metal shielding cover joining shielding structure extends on the minor non-joining side; a second minor terminal module, and the second minor terminal module comprises a second minor signal terminal, a second minor metal shielding cover, and a second minor signal terminal insulating core, and the second minor signal terminal comprises a second minor signal terminal mating structure and a second minor signal terminal joining structure, and the second minor metal shielding cover comprises a second minor metal shielding cover mating shielding structure, a second minor metal shielding cover joining shielding structure, a second minor metal shielding cover mating passage, and a second minor metal shielding cover joining passage, and the second minor signal terminal insulating core is mated with the second minor signal terminal, and the second minor terminal module is configured in the connector insulating core, so that the second minor signal terminal mating structure is located in the connector insulating core mating space and the second minor signal terminal joining structure is located in the connector insulating core joining space, and the second minor metal shielding cover is mated with the second minor signal terminal insulating core, so that the second minor metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the second minor signal terminal mating structure to provide mating via the second minor metal shielding cover mating passage on the second mating side, allowing the second minor metal shielding cover joining passage to be in continuity with the connector insulating core joining space, allowing the second minor signal terminal joining structure to provide joining via the second minor metal shielding cover joining passage on the second joining side, wherein the second minor metal shielding cover mating shielding structure extends on the second non-mating side, and the second minor metal shielding cover joining shielding structure extends on the second non-joining side; and a minor bridging conductive component, and the minor bridging conductive component comprises a minor bridging conductive component non-mating side shielding wall, and the minor bridging conductive component is configured in the connector insulating core, so that the minor bridging conductive component non-mating side shielding wall extends on the minor non-mating side, and is positioned between the first minor terminal module and the second minor terminal module, and further so that the minor bridging conductive component bridges across the first minor signal terminal and the second minor signal terminal, and respectively joins the first minor metal shielding cover and the second minor metal shielding cover, allowing the minor bridging conductive component non-mating side shielding wall, the first minor metal shielding cover mating shielding structure, and the second minor metal shielding cover mating shielding structure to be electrically connected, respectively providing a shielding environment for the first minor signal terminal mating structure and the second minor signal terminal mating structure, allowing the first minor metal shielding cover joining shielding structure and the second minor metal shielding cover joining shielding structure to be electrically connected, providing a shielding environment for the first minor signal terminal joining structure and the second minor signal terminal joining structure.

[0012] Preferably, the board-end connector said above, wherein the first minor metal shielding cover extension shielding structure is located between the first major signal terminal and the first minor signal terminal to provide shielding, and the second minor metal shielding cover extension shielding structure is located between the second major signal terminal and the second minor signal terminal to provide shielding.

[0013] Compared to the prior art, the present application provides a connector assembly and board-end connector thereof, wherein the board-end connector comprises multiple terminal modules, each of which comprises a signal terminal and a metal shielding cover. The metal shielding cover provides a shielding environment for the signal terminal to reduce the level of crosstalk interference when the signal terminal transmits high-speed signals.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The above and other aspects, features and other advantages of the present invention will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0015] FIG. **1** is a three-dimensional schematic view of the connector assembly in an embodiment of the present application.

[0016] FIG. **2** is a three-dimensional schematic view of the connector assembly in an embodiment of the present application.

[0017] FIG. **3** is a three-dimensional schematic view of the connector assembly in an embodiment of the present application.

[0018] FIG. **4** is a three-dimensional schematic view of the connector insulating core in an embodiment of the present application.

[0019] FIG. **5** is a three-dimensional schematic view of the connector insulating core and terminal module in an embodiment of the present application.

[0020] FIG. **6** is a three-dimensional schematic view of the connector insulating core and terminal module in an embodiment of the present application.

[0021] FIG. **7** is a three-dimensional schematic view of the terminal module in an embodiment of the present application.

[0022] FIG. **8** is a three-dimensional schematic view of the terminal module in an embodiment of the present application.

[0023] FIG. **9** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application from a first viewing angle.

[0024] FIG. **10** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application from a second viewing angle.

[0025] FIG. **11** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application from a third viewing angle.

[0026] FIG. **12** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.

[0027] FIG. **13** is a three-dimensional schematic view of the bridging conductive component in an embodiment of the present application.

[0028] FIG. **14** is a three-dimensional schematic view of the board-end connector in an embodiment of the present application.

[0029] FIG. **15** is a three-dimensional schematic view of the board-end connector in an embodiment of the present application.

[0030] FIG. **16** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.

[0031] FIG. **17** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.

[0032] FIG. **18** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.

[0033] FIG. **19** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.
[0034] FIG. **20** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.
[0035] FIG. **21** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.
[0036] FIG. **22** is a three-dimensional schematic view of a partial component of the board-end connector in an embodiment of the present application.
[0037] FIG. **23** is a three-dimensional schematic view of the bridging conductive component in an embodiment of the present application from a first viewing angle.
[0038] FIG. **24** is a three-dimensional schematic view of the bridging conductive component in an embodiment of the present application from a second viewing angle.
[0039] FIG. **25** is a three-dimensional schematic view of the bridging conductive component in an embodiment of the present application.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

[0040] Embodiments of the present invention will now be described in detail with reference to the accompanying drawings. The invention may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art.
[0041] The present application provides a connector assembly and board-end connector thereof, wherein the board-end connector comprises a metal shielding cover capable of providing a shielding environment to reduce the degree of crosstalk interference affecting the signal terminals of the board-end connector when transmitting high-speed signals.
[0042] For a detailed description of the embodiments disclosed in the present application, please refer to FIGS. **1** to **25**.
[0043] In the embodiments shown in FIGS. **1** to **25**, a connector assembly **1** is disclosed, wherein the connector assembly **1** is equipped with a circuit board material **2**. The circuit board material **2** may be, for example, a circuit board or a flexible circuit board and comprises a first major circuit board material signal terminal joining circuit **21**, a second major circuit board material signal terminal joining circuit **22**, a first minor circuit board material signal terminal joining circuit **23**, and a second minor circuit board material signal terminal joining circuit **24**. The connector assembly **1** comprises: a board-end connector **11** and a mating component **12**. The mating component **12** comprises a first major mating component signal terminal mating circuit **121**, a second major mating component signal terminal mating circuit **122**, a first minor mating component signal terminal mating circuit **123**, and a second minor mating component signal terminal mating circuit **124**.
[0044] The board-end connector **11** comprises: a connector insulating core **110**, a first major terminal module **111**, a second major terminal module **112**, and a major bridging conductive component **113**. The connector insulating core **110** comprises a major mating side L**111**, a major non-mating side L**112**, a major joining side L**121**, a major non-joining side L**122**, a first minor mating side L**211**, a first minor non-mating side L**212**, a first minor joining side L**221**, a first minor non-joining side L**222**, a connector insulating core mating space S**1**, and a connector insulating core joining space S**2**. The major non-mating side L**112** and the major mating side L**111** are substantially different, and the major non-joining side L**122** and the major joining side L**121** are substantially different. Additionally, the minor non-mating side L**212** and the minor mating side L**211** are substantially different, and the minor non-joining side L**222** and the minor joining side L**221** are substantially different.
[0045] The first major terminal module **111** comprises a first major signal terminal **1111**, a first major metal shielding cover **1112**, and a first major signal terminal insulating core **1113**. The first

major signal terminal **1111** comprises a first major signal terminal extension structure **11110**, a first major signal terminal mating structure **11111**, and a first major signal terminal joining structure **11112**. The first major signal terminal extension structure **11110** is respectively connected to the first major signal terminal mating structure **11111** and the first major signal terminal joining structure **11112**. The first major metal shielding cover **1112** comprises a first major metal shielding cover extension shielding structure **11120**, a first major metal shielding cover mating shielding structure **11121**, a first major metal shielding cover joining shielding structure **11122**, a first major metal shielding cover mating passage **11123**, and a first major metal shielding cover joining passage **11124**.

[0046] It should be noted that the first major signal terminal insulating core **1113** is mated with the first major signal terminal **1111**. Preferably, the first major signal terminal insulating core **1113** is mated with the first major signal terminal extension structure **11110**, and the first major terminal module **111** is configured in the connector insulating core **110**, such that the first major signal terminal mating structure **11111** is located in the connector insulating core mating space S**1** and the first major signal terminal joining structure **11112** is located in the connector insulating core joining space S**2**. The first major metal shielding cover **1112** is mated with the first major signal terminal insulating core **1113**, such that the first major metal shielding cover mating passage **11123** is in continuity with the connector insulating core mating space S**1**, allowing the first major signal terminal mating structure **11111** to mate via the first major metal shielding cover mating passage **11123** on the major mating side L**111**, and allowing the first major metal shielding cover joining passage **11124** to be in continuity with the connector insulating core joining space S**2**, so that the first major signal terminal joining structure **11112** can join via the first major metal shielding cover joining passage **11124** on the major joining side L**121**, wherein the first major metal shielding cover mating shielding structure **11121** extends on the major non-mating side L**112**, and the first major metal shielding cover joining shielding structure **11122** extends on the major non-joining side L**122**. Preferably, the first major metal shielding cover extension shielding structure **11120** extends on the major mating side L**111** to provide a shielding environment for the first major signal terminal extension structure **11110**.

[0047] The second major terminal module **112** comprises a second major signal terminal **1121**, a second major metal shielding cover **1122**, and a second major signal terminal insulating core **1123**. The second major signal terminal **1121** comprises a second major signal terminal extension structure **11210**, a second major signal terminal mating structure **11211**, and a second major signal terminal joining structure **11212**. The second major signal terminal extension structure **11210** connects the second major signal terminal mating structure **11211** and the second major signal terminal joining structure **11212**. The second major metal shielding cover **1122** comprises a second major metal shielding cover mating shielding structure **11221**, a second major metal shielding cover joining shielding structure **11222**, a second major metal shielding cover mating passage **11223**, and a second major metal shielding cover joining passage **11224**.

[0048] It should be noted that the second major signal terminal insulating core **1123** is mated with the second major signal terminal **1121**. Preferably, the second major signal terminal insulating core **1123** is mated with the second major signal terminal extension structure **11210**, and the second major terminal module **112** is configured in the connector insulating core **110**, such that the second major signal terminal mating structure **11211** is located in the connector insulating core mating space S**1** and the second major signal terminal joining structure **11212** is located in the connector insulating core joining space S**2**. The second major metal shielding cover **1122** is mated with the second major signal terminal insulating core **1123**, such that the second major metal shielding cover mating passage **11223** is in continuity with the connector insulating core mating space S**1**, allowing the second major signal terminal mating structure **11211** to mate via the second major metal shielding cover mating passage **11223** on the major mating side L**111**, and allowing the second major metal shielding cover joining passage **11224** to be in continuity with the connector

insulating core joining space S**2**, so that the second major signal terminal joining structure **11212** can join via the second major metal shielding cover joining passage **11224** on the major joining side L**121**, wherein the second major metal shielding cover mating shielding structure **11221** extends on the major non-mating side L**112**, and the second major metal shielding cover joining shielding structure **11222** extends on the major non-joining side L**122**. Preferably, the second major metal shielding cover extension shielding structure **11220** extends on the major mating side L**111** to provide a shielding environment for the second major signal terminal extension structure **11210**.
[0049] The major bridging conductive component **113** comprises a first major bridging conductive component non-mating side shielding wall **11311**, a major bridging conductive component mating side shielding wall **1132**, and a major bridging conductive component non-mating side abutting wall **1133**. The major bridging conductive component **113** is configured in the connector insulating core **110**, such that the major bridging conductive component mating side shielding wall **1132** respectively provides a shielding environment for the first major signal terminal mating structure **11111** and the second major signal terminal mating structure **11211**, thereby reducing the degree of crosstalk interference during high-speed signal transmission between the first major signal terminal mating structure **11111** and the second major signal terminal mating structure **11211**. Furthermore, the first major bridging conductive component non-mating side shielding wall **11311**, the first major metal shielding cover mating shielding structure **11121**, and the second major metal shielding cover mating shielding structure **11221** are electrically connected to provide transmission of shielded signals, respectively providing shielding environments for the first major signal terminal mating structure **11111** and the second major signal terminal mating structure **11211**, thereby reducing the degree of crosstalk interference during high-speed signal transmission. Additionally, the first major metal shielding cover joining shielding structure **11122** and the second major metal shielding cover joining shielding structure **11222** are electrically connected to provide transmission of shielded signals, respectively providing shielding environments for the first major signal terminal joining structure **11112** and the second major signal terminal joining structure **11212**, thereby reducing the degree of crosstalk interference during high-speed signal transmission.
[0050] It should be noted that, in the above embodiment, the major bridging conductive component **113** spans across the first major signal terminal **1111** and the second major signal terminal **1121**, respectively, and respectively joins with the first major metal shielding cover **1112** and the second major metal shielding cover **1122** to provide transmission of a shielding signal. The first major bridging conductive component non-mating side shielding wall **11311** is positioned between the first major terminal module **111** and the second major terminal module **112**, respectively joining the first major metal shielding cover **1112** and the second major metal shielding cover **1122** to provide transmission of a shielding signal. The first major bridging conductive component non-mating side shielding wall **11311** extends along the major non-mating side L**112**, while the major bridging conductive component mating side shielding wall **1132** extends along the major mating side L**111**. The major bridging conductive component non-mating side abutting wall **1133** extends along the major non-mating side L**112**, and the major bridging conductive component non-mating side abutting wall **1133** optionally features a protruding structure, allowing it to abut the first major metal shielding cover **1112** and the second major metal shielding cover **1122** to provide transmission of a shielding signal.
[0051] It should be noted that, in the present application, the first major terminal module **111** and the second major terminal module **112** are designed to be modular, such that the size specifications of the first major terminal module **111** and the second major terminal module **112** are essentially the same. Thus, the first major terminal module **111** and the second major terminal module **112** can be manufactured using the same equipment or process, thereby reducing the manufacturing cost of the board-end connector **11**.
[0052] Additionally, in the embodiments shown in FIGS. **5** to **6**, the connector insulating core **110** comprises a first major terminal module core configuration structure **1101** and a second major

terminal module core configuration structure **1102**. The first major terminal module core configuration structure **1101** and the second major terminal module core configuration structure **1102** are configuration slot structures, which allow for the respective (sequential) configuration of the first major terminal module **111** and the second major terminal module **112**. It should be noted that, in the present application, the connector insulating core **110** can adjust the quantity of the first major terminal module core configuration structure **1101** and the second major terminal module core configuration structure **1102** to increase or decrease the number of configurations of the first major terminal module **111** and the second major terminal module **112**, thereby ensuring that the electrical characteristics of the board-end connector **11** meet the requirements. Optionally, as shown in FIGS. **3**, **13**, and **23**, the first major bridging conductive component non-mating side shielding wall **11311** can be a bent-type wall, allowing the first major bridging conductive component non-mating side shielding wall **11311** to elastically abut the first major metal shielding cover **1112** and the second major metal shielding cover **1122**, respectively, thereby maintaining the first major bridging conductive component non-mating side shielding wall **11311** between the first major metal shielding cover **1112** and the second major metal shielding cover **1122**.

[0053] Optionally, as shown in FIGS. **19** to **22** and **25**, the first major bridging conductive component non-mating side shielding wall **11311** can be a shear-type wall, allowing the first major bridging conductive component non-mating side shielding wall **11311** to be locked by the first major metal shielding cover **1112** and the second major metal shielding cover **1122** via a concave-convex structure, thereby maintaining the first major bridging conductive component non-mating side shielding wall **11311** between the first major metal shielding cover **1112** and the second major metal shielding cover **1122**.

[0054] In the embodiment shown in FIGS. **3** and **7** to **8**, the board-end connector **11** further comprises a third major terminal module **117**. The third major terminal module **117** comprises a third major signal terminal **1171**, a third major metal shielding cover **1172**, and a third major signal terminal insulating core **1173**. The third major signal terminal **1171** comprises a third major signal terminal mating structure **11711** and a third major signal terminal joining structure **11712**. The third major metal shielding cover **1172** comprises a third major metal shielding cover mating shielding structure **11721**, a third major metal shielding cover joining shielding structure **11722**, a third major metal shielding cover mating passage **11723**, and a third major metal shielding cover joining passage **11724**.

[0055] It should be noted that the third major signal terminal insulating core **1173** is mated with the third major signal terminal **1171**, and the third major terminal module **117** is configured in the connector insulating core **110**, such that the third major signal terminal mating structure **11711** is positioned in the connector insulating core mating space S**1** and the third major signal terminal joining structure **11712** is positioned in the connector insulating core joining space S**2**, respectively. The third major metal shielding cover **1172** is mated with the third major signal terminal insulating core **1173**, such that the third major metal shielding cover mating passage **11723** is in continuity with the connector insulating core mating space S**1**, allowing the third major signal terminal mating structure **11711** to mate via the third major metal shielding cover mating passage **11723** on the major mating side L**111**. The third major metal shielding cover joining passage **11724** is in continuity with the connector insulating core joining space S**2**, allowing the third major signal terminal joining structure **11312** to join via the third major metal shielding cover joining passage **11724** on the major joining side L**121**. The third major metal shielding cover mating shielding structure **11721** extends along the major non-mating side L**112**, while the third major metal shielding cover joining shielding structure **11722** extends along the major non-joining side L**122**.

[0056] Accordingly, the major bridging conductive component **113** also comprises a second major bridging conductive component non-mating side shielding wall **11312**. The second major bridging conductive component non-mating side shielding wall **11312** extends from the major non-mating side L**112**, and is located between the second major terminal module **112** and the third major terminal module **113**. The major bridging conductive component **113** further bridges over the second major signal terminal **1121** and the third major signal terminal **1131**, respectively, mating with the second major metal shielding cover **1122** and the third major metal shielding cover **1132**, allowing the second major bridging conductive component non-mating side shielding wall **11312**, the second major metal shielding cover mating shielding structure **11221**, and the third major metal shielding cover mating shielding structure **11721** to be electrically connected, providing a shielding environment for the second major signal terminal mating structure **11211** and the third major signal terminal mating structure **11711**, reducing crosstalk interference when the second major signal terminal mating structure **11211** and the third major signal terminal mating structure **11711** transmits high-speed signals. Furthermore, the second major metal shielding cover joining shielding structure **11222** and the third major metal shielding cover joining shielding structure **11722** are electrically connected, providing a shielding environment for the second major signal terminal joining structure **11212** and the third major signal terminal joining structure **11712**, reducing crosstalk interference when the second major signal terminal joining structure **11212** and the third major signal terminal joining structure **11712** transmits high-speed signals.

[0057] It should be noted that in the embodiments shown in FIGS. **23** to **25**, the major bridging conductive component **113** omits the second major bridging conductive component non-mating side shielding wall **11312**, and only comprises the first major bridging conductive component non-mating side shielding wall **11311**. This allows for a reduction in the size of the major bridging conductive component **113** to meet usage requirements. Additionally, in this application, the major bridging conductive component **113** may comprise multiple first major bridging conductive component non-mating side shielding walls **11311** to accommodate the usage requirements of increased size of the major bridging conductive component **113**, and is not limited to the embodiment disclosed above. The first minor terminal module **114** comprises a first minor signal terminal **1141**, a first minor metal shielding cover **1142**, and a first minor signal terminal insulating core **1143**. The first minor signal terminal **1141** comprises a first minor signal terminal extension structure **11410**, a first minor signal terminal mating structure **11411**, and a first minor signal terminal joining structure **11412**. The first minor signal terminal extension structure **11410** respectively connects the first minor signal terminal mating structure **11411** and the first minor signal terminal joining structure **11412**. The first minor metal shielding cover **1142** comprises a first minor metal shielding cover extension shielding structure **11420**, a first minor metal shielding cover mating shielding structure **11421**, a first minor metal shielding cover joining shielding structure **11422**, a first minor metal shielding cover mating passage **11423**, and a first major metal shielding cover joining passage **11424**.

[0058] It should be noted that the first minor signal terminal insulating core **1143** is mated with the first minor signal terminal **1141**. Preferably, the first minor signal terminal insulating core **1143** is mated with the first minor signal terminal extension structure **11410**, and the first minor terminal module **114** is configured in the connector insulating core **110**, positioning the first minor signal terminal mating structure **11411** within the connector insulating core mating space S**1**, and positioning the first minor signal terminal joining structure **11412** within the connector insulating core joining space S**2**, respectively. The first minor metal shielding cover **1142** is mated with the first minor signal terminal insulating core **1143**, allowing the first minor metal shielding cover mating passage **11423** to be in continuity with the connector insulating core mating space S**1**, enabling the first minor signal terminal mating structure **11411** to provide mating on the minor mating side L**211** via the first minor metal shielding cover mating passage **11423**. Additionally, the first major metal shielding cover joining passage **11424** is in continuity with the connector insulating core joining space S**2**, allowing the first minor signal terminal joining structure **11412** to provide joining on the minor joining side L**221** via the first major metal shielding cover joining passage **11424**. The first minor metal shielding cover mating shielding structure **11421** extends from the minor non-mating side L**212**, while the first minor metal shielding cover joining shielding structure **11422** extends from the minor non-joining side L**222**. Preferably, the first minor metal shielding cover extension shielding structure **11420** extends from the minor mating side L**211**, providing a shielding environment for the first minor signal terminal extension structure **11410**.

[0059] The second minor terminal module **115** comprises a second minor signal terminal **1151**, a second minor metal shielding cover **1152**, and a second minor signal terminal insulating core **1153**. The second minor signal terminal **1151** comprises a second minor signal terminal extension structure **11510**, a second minor signal terminal mating structure **11511**, and a second minor signal terminal joining structure **11512**. The second minor signal terminal extension structure **11510** respectively connects the second minor signal terminal mating structure **11511** and the second minor signal terminal joining structure **11512**. The second minor metal shielding cover **1152** comprises a second minor metal shielding cover extension shielding structure **11520**, a second minor metal shielding cover mating shielding structure **11521**, a second minor metal shielding cover joining shielding structure **11522**, a second minor metal shielding cover mating passage **11523**, and a second minor metal shielding cover joining passage **11524**.

[0060] It should be noted that the second minor signal terminal insulating core **1153** is mated with the second minor signal terminal **1151**. Preferably, the second minor signal terminal insulating core **1153** is mated with the second minor signal terminal extension structure **11510**. The second minor terminal module **115** is assembled within the connector insulating core **110**, such that the second minor signal terminal mating structure **11511** is located within the connector insulating core mating space S**1** and the second minor signal terminal joining structure **11512** is located within the connector insulating core joining space S**2**. The second minor metal shielding cover **1152** is mated with the second minor signal terminal insulating core **1153**, such that the second minor metal shielding cover mating passage **11523** is in continuity with the connector insulating core mating space S**1**, allowing the second minor signal terminal mating structure **11511** to mate via the second minor metal shielding cover mating passage **11523** on the minor mating side L**211**. The second minor metal shielding cover joining passage **11524** is in continuity with the connector insulating core joining space S**2**, allowing the second minor signal terminal joining structure **11512** to join via the second minor metal shielding cover joining passage **11524** on the minor joining side L**221**. The second minor metal shielding cover mating shielding structure **11521** extends on the minor non-mating side L**212**, and the second minor metal shielding cover joining shielding structure **11522** extends on the minor non-joining side L**222**. Preferably, the second minor metal shielding cover extension shielding structure **11520** extends on the minor mating side L**211** to provide a shielding environment for the second minor signal terminal extension structure **11510**.

[0061] The minor bridging conductive component **116** comprises the minor bridging conductive component non-mating side shielding wall **1161**, the minor bridging conductive component mating side shielding wall **1162**, and the minor bridging conductive component non-mating side abutting wall **1163**. The minor bridging conductive component **116** is configured within the connector insulating core **110**, such that the minor bridging conductive component mating side shielding wall **1162** provides a shielding environment for the first minor signal terminal mating structure **11411** and the second minor signal terminal mating structure **11511**, reducing the level of crosstalk interference when the first minor signal terminal mating structure **11411** and the second minor signal terminal mating structure **11511** transmit high-speed signals. The minor bridging conductive component non-mating side shielding wall **1161**, the first minor metal shielding cover mating shielding structure **11421**, and the second minor metal shielding cover mating shielding structure **11521** are electrically connected to provide transmission of a shielding signal and they provide a shielding environment for the first minor signal terminal mating structure **11411** and the second minor signal terminal mating structure **11511**, reducing the level of crosstalk interference when the first minor signal terminal mating structure **11411** and the second minor signal terminal mating structure **11511** transmit high-speed signals Additionally, the first minor metal shielding cover joining shielding structure **11422** and the second minor metal shielding cover joining shielding structure **11522** are electrically connected to provide transmission of a shielding signal, providing a shielding environment for the first minor signal terminal joining structure **11412** and the second minor signal terminal joining structure **11512**, thereby reducing the level of crosstalk interference when the first minor signal terminal joining structure **11412** and the second minor signal terminal joining structure **11512** transmit high-speed signals.

[0062] It should be noted that in the above embodiment, the minor bridging conductive component **116** bridges over the first minor signal terminal **1141** and the second minor signal terminal **1151**, and respectively joins the first minor metal shielding cover **1142** and the second minor metal shielding cover **1152** to provide a transmission of a shielding signal. The minor bridging conductive component non-mating side shielding wall **1161** is located between the first minor terminal module **114** and the second minor terminal module **115**, and respectively joins the first minor metal shielding cover **1142** and the second minor metal shielding cover **1152** to provide a transmission of a shielding signal. The minor bridging conductive component non-mating side shielding wall **1161** extends on the minor non-mating side L**212**, and the minor bridging conductive component mating side shielding wall **1162** extends on the minor mating side L**211**. The minor bridging conductive component non-mating side abutting wall **1163** extends on the minor non-mating side L**212**, and respectively abuts the first minor metal shielding cover **1142** and the second minor metal shielding cover **1152** to provide a transmission of a shielding signal.

[0063] Additionally, the first minor metal shielding cover extension shielding structure **11420** is located between the first major signal terminal **1111** and the first minor signal terminal **1141** to provide shielding, and the second minor metal shielding cover extension shielding structure **11520** is located between the second major signal terminal **1121** and the second minor signal terminal **1151** to provide shielding.

[0064] In the embodiments shown in FIGS. **1** to **3** and FIGS. **14** to **15**, the board-end connector **11** is capable of being joined with the circuit board **2**, such that the first major signal terminal joining structure **11112** can be joined with the first major circuit board signal terminal mating circuit **21** on the major mating side L**121** and the second major signal terminal joining structure **11212** can be joined with the second major circuit board signal terminal mating circuit **22** on the major mating side L**121** to transmit electrical signals. Additionally, the first minor signal terminal joining structure **11412** can be joined with the first minor circuit board signal terminal mating circuit **23** on the minor mating side L**221** and the second minor signal terminal joining structure **11512** can be joined with the second minor circuit board signal terminal mating circuit **24** on the minor mating side L**221** to transmit electrical signals.

[0065] Furthermore, the mating component **12**, such as a PCIE card, can mate with the board-end connector **11**, allowing the first major signal terminal mating structure **11111** to mate with the first major mating component signal terminal mating circuit **121** on the major mating side L**111** and allowing the second major signal terminal mating structure **11211** to mate with the second major mating component signal terminal mating circuit **122** on the major mating side L**111** to transmit electrical signals. Additionally, the first minor signal terminal mating structure **11411** can mate with the first minor mating component signal terminal mating circuit **123** on the minor mating side L**211**, and the second minor signal terminal mating structure **11511** can mate with the second minor mating component signal terminal mating circuit **124** on the minor mating side L**211** to transmit electrical signals.

[0066] It should be noted that the connector assembly and board-end connector thereof in this application can omit certain components and are not limited to the above embodiments.

[0067] In conclusion, this application provides a connector assembly and board-end connector thereof, wherein the board-end connector comprises multiple terminal modules, each of which comprises a signal terminal and a metal shielding cover. The metal shielding cover provides a shielding environment for the signal terminal to reduce the degree of crosstalk interference when the signal terminal transmits high-speed signals.

[0068] The examples above are only illustrative to explain principles and effects of the invention, but not to limit the invention. It will be apparent to those skilled in the art that modifications and variations can be made without departing from the spirit and scope of the invention. Therefore, the protection range of the rights of the invention should be as defined by the appended claims.

Claims

1. A board-end connector, the board-end connector comprising: a connector insulating core, the connector insulating core comprising a major mating side, a major non-mating side, a major joining side, a major non-joining side, a connector insulating core mating space, and a connector insulating core joining space, wherein the major non-mating side and the major mating side are substantially different, and the major non-joining side and the major joining side are substantially different; a first major terminal module, the first major terminal module comprising a first major signal terminal, a first major metal shielding cover, and a first major signal terminal insulating core, the first major signal terminal comprising a first major signal terminal mating structure and a first major signal terminal joining structure, the first major metal shielding cover comprising a first major metal shielding cover mating shielding structure, a first major metal shielding cover joining shielding structure, a first major metal shielding cover mating passage, and a first major metal shielding cover joining passage, the first major signal terminal insulating core being mated with the first major signal terminal, and the first major terminal module being configured in the connector insulating core, such that the first major signal terminal mating structure is located in the connector insulating core mating space and the first major signal terminal joining structure is located in the connector insulating core joining space, the first major metal shielding cover being mated with the first major signal terminal insulating core, such that the first major metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the first major signal terminal mating structure to be mated via the first major metal shielding cover mating passage on the major mating side, and the first major metal shielding cover joining passage is in continuity with the connector insulating core joining space, allowing the first major signal terminal joining structure to be joined via the first major metal shielding cover joining passage on the major joining side, wherein the first major metal shielding cover mating shielding structure extends on the major non-mating side, and the first major metal shielding cover joining shielding structure extends on the major non-joining side; a second major terminal module, the second major terminal module comprising a second major signal terminal, a second major metal shielding cover, and a second major signal terminal insulating core, the second major signal terminal comprising a second major signal terminal mating structure and a second major signal terminal joining structure, the second major metal shielding cover comprising a second major metal shielding cover mating shielding structure, a second major metal shielding cover joining shielding structure, a second major metal shielding cover mating passage, and a second major metal shielding cover joining passage, the second major signal terminal insulating core being mated with the second major signal terminal, and the second major terminal module being configured in the connector insulating core, such that the second major signal terminal mating structure is located in the connector insulating core mating space and the second major signal terminal joining structure is located in the connector insulating core joining space, the second major metal shielding cover being mated with the second major signal terminal insulating core, such that the second major metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the second major signal terminal mating structure to be mated via the second major metal shielding cover mating passage on the major mating side, and the second major metal shielding cover joining passage is in continuity with the connector insulating core joining space, allowing the second major signal terminal joining structure to be joined via the second major metal shielding cover joining passage on the major joining side, wherein the second major metal shielding cover mating shielding structure extends on the major non-mating side, and the second major metal shielding cover joining shielding structure extends on the major non-joining side; and a major bridging conductive component, the major bridging conductive component comprising a first major bridging conductive component non-mating side shielding wall, the major bridging conductive component being configured in the connector insulating core, such that the first major bridging conductive component non-mating side shielding wall extends on the major non-mating side and is located between the first major terminal module and the second major terminal module, and the major bridging conductive component bridging over the first major signal terminal and the second major signal terminal, respectively joining the first major metal shielding cover and the second major metal shielding cover, allowing the first major bridging conductive component non-mating side shielding wall, the first major metal shielding cover mating shielding structure, and the second major metal shielding cover mating shielding structure to be electrically connected, providing a shielding environment for the first major signal terminal mating structure and the second major signal terminal mating structure, and allowing the first major metal shielding cover joining shielding structure and the second major metal shielding cover joining shielding structure to be electrically connected, providing a shielding environment for the first major signal terminal joining structure and the second major signal terminal joining structure.

2. The board-end connector of claim 1, wherein the first major terminal module and the second major terminal module are substantially the same; the connector insulating core comprises a first major terminal module core configuration structure and a second major terminal module core configuration structure, and the first major terminal module core configuration structure is configured to the first major terminal module, and the second major terminal module core configuration structure is configured to the second major terminal module.

3. The board-end connector of claim 1, further comprising: a third major terminal module, the third major terminal module comprising a third major signal terminal, a third major metal shielding cover, and a third major signal terminal insulating core, and the third major signal terminal comprises a third major signal terminal mating structure and a third major signal terminal joining structure, and the third major metal shielding cover comprises a third major metal shielding cover mating shielding structure, a third major metal shielding cover joining shielding structure, a third major metal shielding cover mating passage, and a third major metal shielding cover joining passage, and the third major signal terminal insulating core is mated with the third major signal terminal, and the third major terminal module is configured in the connector insulating core, positioning the third major signal terminal mating structure within the connector insulating core mating space, and positioning the third major signal terminal joining structure within the connector insulating core joining space, and the third major metal shielding cover is mated with the third major signal terminal insulating core, so that the third major metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the third major signal terminal mating structure to mate via the third major metal shielding cover mating passage on the major mating side, and the third major metal shielding cover joining passage is in continuity with the connector insulating core joining space, allowing the third major signal terminal joining structure to join via the third major metal shielding cover joining passage on the major joining side, wherein the third major metal shielding cover mating shielding structure extends on the major non-mating side, and the third major metal shielding cover joining shielding structure extends on the major non-joining side; wherein the major bridging conductive component further comprises a second major bridging conductive component non-mating side shielding wall, and the second major bridging conductive component non-mating side shielding wall extends on the major non-mating side and is located between the second major terminal module and the third major terminal module. The major bridging conductive component also bridges across the second major signal terminal and the third major signal terminal, respectively joining the second major metal shielding cover and the third major metal shielding cover, thereby allowing the second major bridging conductive component non-mating side shielding wall, the second major metal shielding cover mating shielding structure, and the third major metal shielding cover mating shielding structure to be electrically connected, providing a shielding environment for the second major signal terminal mating structure and the third major signal terminal mating structure.

4. The board-end connector of claim 1, wherein the major bridging conductive component further comprises a major bridging conductive component mating side shielding wall and a major bridging conductive component non-mating side abutting wall; wherein the major bridging conductive component mating side shielding wall extends on the major mating side, providing a shielding environment for the first major signal terminal mating structure and the second major signal terminal mating structure; and the major bridging conductive component non-mating side abutting wall extends on the major non-mating side, respectively abutting the first major metal shielding cover and the second major metal shielding cover.

5. The board-end connector of claim 1, wherein the first major bridging conductive component non-mating side shielding wall is a bent type wall, allowing the first major bridging conductive component non-mating side shielding wall to elastically abut the first major metal shielding cover and the second major metal shielding cover, respectively.

6. The board-end connector of claim 1, wherein the first major bridging conductive component non-mating side shielding wall is a shear type wall, allowing the first major bridging conductive component non-mating side shielding wall to lock the first major metal shielding cover and the second major metal shielding cover.

7. The board-end connector of claim 1, wherein the connector insulating core further comprises a minor mating side, a minor non-mating side, a minor joining side, and a minor non-joining side, wherein the minor non-mating side and the minor mating side are substantially different, and the minor non-joining side and the minor joining side are substantially different; and the board-end connector further comprises: a first minor terminal module, and the first minor terminal module comprises a first minor signal terminal, a first minor metal shielding cover, and a first minor signal terminal insulating core, and the first minor signal terminal comprises a first minor signal terminal mating structure and a first minor signal terminal joining structure, and the first minor metal shielding cover comprises a first minor metal shielding cover mating shielding structure, a first minor metal shielding cover joining shielding structure, a first minor metal shielding cover mating passage, and a first major metal shielding cover joining passage, and the first minor signal terminal insulating core is mated with the first minor signal terminal, and the first minor terminal module is configured in the connector insulating core, so that the first minor signal terminal mating structure is located in the connector insulating core mating space and the first minor signal terminal joining structure is located in the connector insulating core joining space, and the first minor metal shielding cover is mated with the first minor signal terminal insulating core, so that the first minor metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the first minor signal terminal mating structure to provide mating via the first minor metal shielding cover mating passage on the minor mating side, allowing the first major metal shielding cover joining passage to be in continuity with the connector insulating core joining space, allowing the first minor signal terminal joining structure to provide joining via the first major metal shielding cover joining passage on the minor joining side, wherein the first minor metal shielding cover mating shielding structure extends on the minor non-mating side, and the first minor metal shielding cover joining shielding structure extends on the minor non-joining side; a second minor terminal module, and the second minor terminal module comprises a second minor signal terminal, a second minor metal shielding cover, and a second minor signal terminal insulating core, and the second minor signal terminal comprises a second minor signal terminal mating structure and a second minor signal terminal joining structure, and the second minor metal shielding cover comprises a second minor metal shielding cover mating shielding structure, a second minor metal shielding cover joining shielding structure, a second minor metal shielding cover mating passage, and a second minor metal shielding cover joining passage, and the second minor signal terminal insulating core is mated with the second minor signal terminal, and the second minor terminal module is configured in the connector insulating core, so that the second minor signal terminal mating structure is located in the connector insulating core mating space and the second minor signal terminal joining structure is located in the connector insulating core joining space, and the second minor metal shielding cover is mated with the second minor signal terminal insulating core, so that the second minor metal shielding cover mating passage is in continuity with the connector insulating core mating space, allowing the second minor signal terminal mating structure to provide mating via the second minor metal shielding cover mating passage on the second mating side, allowing the second minor metal shielding cover joining passage to be in continuity with the connector insulating core joining space, allowing the second minor signal terminal joining structure to provide joining via the second minor metal shielding cover joining passage on the second joining side, wherein the second minor metal shielding cover mating shielding structure extends on the second non-mating side, and the second minor metal shielding cover joining shielding structure extends on the second non-joining side; and a minor bridging conductive component, and the minor bridging conductive component comprises a minor bridging conductive component non-mating side shielding wall, and the minor bridging conductive component is configured in the connector insulating core, so that the minor bridging conductive component non-mating side shielding wall extends on the minor non-mating side, and is positioned between the first minor terminal module and the second minor terminal module, and further so that the minor bridging conductive component bridges across the first minor signal terminal and the second minor signal terminal, and respectively joins the first minor metal shielding cover and the second minor metal shielding cover, allowing the minor bridging conductive component non-mating side shielding wall, the first minor metal shielding cover mating shielding structure, and the second minor metal shielding cover mating shielding structure to be electrically connected, respectively providing a shielding environment for the first minor signal terminal mating structure and the second minor signal terminal mating structure, allowing the first minor metal shielding cover joining shielding structure and the second minor metal shielding cover joining shielding structure to be electrically connected, providing a shielding environment for the first minor signal terminal joining structure and the second minor signal terminal joining structure.

8. The board-end connector of claim 7, wherein the first minor metal shielding cover extension shielding structure is located between the first major signal terminal and the first minor signal terminal to provide shielding, and the second minor metal shielding cover extension shielding structure is located between the second major signal terminal and the second minor signal terminal to provide shielding.